Department of Computer Science and Engineering
University of Dhaka
2nd Year 2nd Semester 2024, Lab Project
Course Code: CSE-2201, Database Management Systems - I

Part I (Design, Implementation and Report Writing)

Consider a relational database model of your own and go for the following tasks.

1. Consider the database of reasonable number of schemas/tables
2. Plan for attributes that cover all general data types
3. Plan for constraints of different types (primary keys, foreign keys, unique keys, check, not null etc.)
4. Create appropriate data for the designed relations
5. Implement the database in Oracle or any DBMS system
6. Plan queries (query statement) and find the answer (at least 15)
7. Plan for non-trivial (canonical cover type) functional dependencies applicable for the schema design.
8. Formulate everything in a **report** which includes:
    a. Brief description of the database system that you are going to implement in the database
    b. Mention schemas with attributes
    c. Show **Schema diagram** of the database
    d. Show **E-R diagram** of the database
    e. Snapshots of **SQL DDL** of all the schemas/tables
    f. Snapshots of the **instances** (data of the populated tables)
    g. **Query statement, Relational Algebra Expression, SQL statement** and snapshots of the **outputs**. SQL statements **should** contain the following :
        i. **natural join**, cross product, **outer join**, **join** with **using**, **on**
        ii. nested sub-query with clauses **some, all, any, exists, unique** etc.
        iii. nested sub-query in **from, where** and **select** (scalar sub-query) clause
        iv. **order by**, **group by**, **having** clauses
        v. Use of **with** clause
        vi. String, set operation
        vii. Update, delete operations
        viii. Use of built-in aggregate function and other important functions
    h. Create views and use those views in answering queries.
    i. Find the list of non-trivial FDs and proof that the schemas are in desired normal forms
    j. Frontend designing tools and techniques, advantages and disadvantages of the implementation
    k. Conclusion of the work

Part II (Web Implementation)

Create a web application based on your designed database.

Part III (Demonstration and Presentation)

1. Create a presentation of the project (5 Minutes)
2. Demonstrate your project (5-8 Minutes)